ANALYSIS REPORT

Repository Status Wave Track Synthesis

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The repository is in an active but stable post-`Wave 2.3` state. No training campaign is currently prepared or running, and `doc/running/active_training_campaign.yaml` has no protected files.

The current scalar program winner is

Surface	Run	Family	Test MAE [deg]	Test RMSE [deg]
`Bw`	`te_periodic_gru_sequence_remote_Bw`	`periodic_gru_sequence_bw`	0.002344	0.002747

That scalar winner is not the whole decision anymore. Commit `b73220679410276246421b7e2832d8878cff90a0` moved promotion toward full `TE Curve Verification Pipeline` curve-following quality.

Current `TE Curve Verification Pipeline` decision

- accepted paper-derived baselines remain `paper_retuned_best_Fw` and `paper_retuned_best_Bw`;
- `tree` remains the strongest repository-owned static baseline;
- `Wave 2.2` is the strongest repository-owned neural branch, while `Wave 2.3` is verified only as an exploratory residual harmonic temporal baseline.

Repository State

The program has closed the following major surfaces:

Surface	Current State	Operational Meaning
`RCIM Model-Bank Reproduction`	closed	faithful full-bank RCIM paper reproduction surface
`Wave 1`	closed	structured static repository baselines
`TE Curve Verification Pipeline`	closed through `Wave 2.3`	official offline curve-verification matrix
`Wave 2.1`	completed	first temporal sequence baselines
`Wave 2.2`	completed	harmonic temporal hybrids, current strongest neural branch
`Wave 2.3`	completed	residual harmonic temporal hybrids, exploratory baseline

The repository now has `53` implemented and benchmarked family surfaces in the master summary. The next active work should not be another blind family addition. The current focus is standardizing curve-first reranking on the expanded `TE Curve Verification Pipeline` metric surface.

Operationally, the repository is ready for decision-oriented analysis rather than another bookkeeping pass. The available evidence now covers scalar leaderboards, official curve reports, direction-aware comparisons, and deployment plausibility for the strongest branches.

Wave 1 Static Baselines

`Wave 1` remains the closed structured-static baseline wave. It covers direction-separated `global`, `Fw`, and `Bw` surfaces for tree, feedforward, periodic MLP, harmonic regression, and residual harmonic MLP families.

The scalar HPO closeout leader is

Rank	Family	Scope	Engine	Test MAE [deg]	Test RMSE [deg]
1	`tree_fw`	`forward`	`bounded_grid`	0.002743	0.003409
2	`tree`	`global`	`bounded_grid`	0.002782	0.003520
3	`tree_bw`	`backward`	`bounded_grid`	0.002954	0.003749

The post-closeout harmonic tracking campaigns improved the harmonic-specific families but did not displace the tree baseline as the strongest `Wave 1` scalar and `TE Curve Verification Pipeline` static family.

Best current harmonic-oriented `Wave 1` scalar results:

Family Surface	Best Run	Harmonic Basis	Test MAE [deg]
`harmonic_regression_fw`	`te_harmonic_dense360_tracking_Fw`	dense `0..360`	0.002916
`residual_harmonic_mlp_bw`	`te_residual_harmonic_rcim_sparse_tracking_Bw`	sparse `RCIM`	0.003042
`periodic_mlp_fw`	`te_periodic_mlp_dense240_tracking_Fw`	dense `0..240`	0.003055
`residual_harmonic_mlp_fw`	`te_residual_harmonic_rcim_sparse_tracking_Fw`	sparse `RCIM`	0.003089

Interpretation

- dense harmonics can help simple harmonic regression in forward-only scalar training;
- sparse `RCIM` remains strong for residual harmonic MLP surfaces;
- `periodic_mlp_harmonic_*` is a report/candidate label for explicit-harmonic `periodic_mlp` candidates, not a separate architecture family;
- `tree` remains the strongest repository-owned static `TE Curve Verification Pipeline` baseline.

Wave 2.1 Temporal Sequence Models

`Wave 2.1` introduced temporal sequence baselines

- `temporal_convolution`;
- `gru_sequence`;
- `lstm_sequence`.

The entry campaign completed all `9` planned runs. Its scalar campaign winner was:

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
`te_gru_sequence_remote_Fw`	`gru_sequence_fw`	`Fw`	0.003333	0.003881

The best global and backward temporal-entry surfaces were:

Surface	Run	Family	Test MAE [deg]
`global`	`te_lstm_sequence_remote_global`	`lstm_sequence`	0.003482
`Bw`	`te_lstm_sequence_remote_Bw`	`lstm_sequence_bw`	0.003557

`TE Curve Verification Pipeline` verified these models as useful exploratory temporal baselines, but plain temporal recurrence did not improve enough over the static tree and paper-derived reference candidates.

Wave 2.2 Harmonic Temporal Hybrids

`Wave 2.2` added explicit periodic harmonic features to temporal convolution, `GRU`, and `LSTM` sequence windows. The first tier used the sparse `RCIM` harmonic list:

`text [0, 1, 3, 39, 40, 78, 81, 156, 162, 240]`

The campaign completed all `9` planned runs. Its scalar winner is also the current program scalar winner:

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
`te_periodic_gru_sequence_remote_Bw`	`periodic_gru_sequence_bw`	`Bw`	0.002344	0.002747

The strongest bidirectional neural candidate is

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
`te_periodic_gru_sequence_remote_global`	`periodic_gru_sequence`	`global`	0.002681	0.002971

`TE Curve Verification Pipeline` confirms the same qualitative result. The official visual and metric reports identify:

Candidate	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error [%]
`periodic_gru_sequence_Bw`	`Bw`	0.002392	0.002639	5.466
`periodic_gru_sequence_global`	`global`	0.002704	0.002949	6.139
`periodic_lstm_sequence_global`	`global`	0.002707	0.002958	6.120

Interpretation

- adding explicit sparse harmonic features to recurrent temporal models is the most successful neural move so far;
- `GRU` and `LSTM` are nearly tied in global scalar and curve metrics, but the backward-only periodic `GRU` is the clearest winner;
- the result is still less deployability-transparent than static tree or harmonic models, so the next decision must be curve-first and TwinCAT-aware.

Wave 2.3 Residual Harmonic Temporal Hybrids

`Wave 2.3` tested a stronger decomposition

- structured harmonic branch for the base TE shape;
- recurrent temporal branch for sequence-local residual correction;
- final prediction as the sum of the two branches.

  The campaign tested `18` runs:
- `residual_harmonic_gru_sequence`;
- `residual_harmonic_lstm_sequence`;

- `global`, `Fw`, and `Bw` surfaces;
- sparse `RCIM`, dense `0..240`, and dense `0..360` harmonic banks.

  The scalar campaign winner was:

Run	Family	Basis	Scope	Test MAE [deg]	Test RMSE [deg]
`te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim`	`residual_harmonic_gru_sequence_fw_sparse_rcim`	sparse `RCIM`	`Fw`	0.003200	0.003635

The scalar leaderboard showed dense variants close to sparse variants in some forward-only pointwise metrics. `TE Curve Verification Pipeline` changed the interpretation: when the models are judged on full curves, dense residual harmonic banks are clearly weaker.

Strongest `Wave 2.3` `TE Curve Verification Pipeline` candidates:

Scope	Candidate	Basis	Curve MAE [deg]	Curve RMSE [deg]	Mean Error [%]
`Fw`	`residual_harmonic_gru_sequence_sparse_rcim_Fw`	sparse `RCIM`	0.003194	0.003499	7.083
`Bw`	`residual_harmonic_lstm_sequence_sparse_rcim_Bw`	sparse `RCIM`	0.003440	0.003793	7.510
`global`	`residual_harmonic_lstm_sequence_sparse_rcim_global`	sparse `RCIM`	0.003368	0.003719	7.409

Dense `Wave 2.3` `TE Curve Verification Pipeline` examples:

Scope	Candidate	Basis	Curve MAE [deg]	Mean Error [%]
`Fw`	`residual_harmonic_gru_sequence_dense240_Fw`	dense `0..240`	0.006983	15.722
`Fw`	`residual_harmonic_gru_sequence_dense360_Fw`	dense `0..360`	0.007869	17.740
`Bw`	`residual_harmonic_lstm_sequence_dense240_Bw`	dense `0..240`	0.007367	16.660
`Bw`	`residual_harmonic_lstm_sequence_dense360_Bw`	dense `0..360`	0.010268	23.355
`global`	`residual_harmonic_lstm_sequence_dense240_global`	dense `0..240`	0.006419	14.460
`global`	`residual_harmonic_lstm_sequence_dense360_global`	dense `0..360`	0.008810	19.916

Interpretation

- sparse `RCIM` is the correct `Wave 2.3` harmonic tier;
- dense residual harmonic expansion overfits or destabilizes the curve surface;
- residual decomposition is useful as a diagnostic branch, but it does not beat

  the simpler `Wave 2.2` periodic recurrent formulation;

- `Wave 2.3` should stay closed as a verified exploratory baseline.

TE Curve Verification Pipeline Current Result

The official `TE Curve Verification Pipeline` package now contains `111` candidates. It evaluates direction-valid held-out TE curves and reports curve-level metrics separately from scalar training metrics.

Current official leaders

Scope	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]
forward overall	`rcim_retuned_GBM19_Fw`	0.001089	0.001299	2.372
backward overall	`periodic_gru_sequence_Bw`	0.002392	0.002639	5.466
paper-derived forward	`paper_retuned_best_Fw`	0.001839	0.002041	4.109
paper-derived backward	`paper_retuned_best_Bw`	0.003675	0.004284	7.572
repository static backward	`tree_Bw`	0.003258	0.003651	7.051
repository global static	`tree_global`	0.003144	0.003533	6.854
repository global neural	`periodic_gru_sequence_global`	0.002704	0.002949	6.139

The important split is

- the recovered/retuned paper reference still owns the best forward curve result;
- `Wave 2.2` owns the best repository neural backward and global neural result;
- `Wave 1` tree remains the most robust repository-owned static baseline;
- `Wave 2.3` adds evidence about harmonic-bank choice, not a new promotion.

Harmonic Bank Conclusions

The harmonic-bank evidence is direction- and family-dependent. There is no single universal winner across every algorithm.

Context	Best Observed Choice	Evidence
`Wave 1` harmonic regression, `Fw`	dense `0..360`	`test_mae = 0.002916`
`Wave 1` periodic MLP, `Fw`	dense `0..240`	`test_mae = 0.003055`
`Wave 1` residual harmonic MLP, `Bw`	sparse `RCIM`	`test_mae = 0.003042`
`Wave 2.2` periodic recurrent	sparse `RCIM` feature tier	`periodic_gru_sequence_Bw`, `test_mae = 0.002344`
`Wave 2.3` residual harmonic temporal	sparse `RCIM`	best scalar and all best `TE Curve Verification Pipeline` surfaces
`Wave 2.3` dense residual temporal	dense `0..240` or `0..360`	poor `TE Curve Verification Pipeline` curve metrics

Practical conclusion

- dense harmonic banks are useful as targeted stress tests and can help some static or low-capacity harmonic models;
- sparse `RCIM` harmonics are the better default for neural hybrid and residual temporal work;

- dense `0..240` and `0..360` should not be promoted into new temporal residual campaigns without a curve-first reason.

Perspective Shift From Commit b73220679410276246421b7e2832d8878cff90a0

Commit `b73220679410276246421b7e2832d8878cff90a0` is titled

```text
Define curve-first TE training strategy
```

Its change of vision is important. Before that commit, the repository could be read as mostly scalar-registry driven: a lower `test_mae` moved the program winner. After that commit, scalar error remains a sanity gate, but not the final promotion rule for TE compensation.

The new interpretation is

1. TE compensation is a continuous curve-following problem.
2. Full held-out curves are evaluation units, not future inputs supplied to the runtime model.
3. Future deployed models must remain causal: current operating state, optional short past history, or causal derived features only.
4. `TE Curve Verification Pipeline` curve metrics and visual overlays become the promotion surface.
5. Scalar winner, curve-first winner, and deployment-ready candidate should be reported separately until `Track 3` online evidence exists. This is a useful correction. It prevents the next work from chasing a small pointwise `MAE` gain that may smooth out oscillations, shift harmonic phase, or fail under repeated-revolution compensation.

Future Development Plan

Step 1: CVP 1.1 Curve-First Reranking

Open the already planned `CVP 1.1 Curve-First Reranking` branch before any new training campaign.

Deliverables

- expanded per-curve metric CSV;
- direction-separated reranking for `Wave 1`, `Wave 2.1`, `Wave 2.2`, and `Wave 2.3`;
- harmonic amplitude and harmonic phase diagnostics on the sparse `RCIM` harmonic set;
- P95 and worst-condition percentage-error tables;
- updated visual overlays if the screened candidate set changes;
- master-summary update separating scalar best from curve-first best.

Step 2: Decide Whether To Retrain Or Promote

If reranking confirms that `Wave 2.2` periodic recurrent models are also the best curve-first neural branch, keep them as the neural reference and improve checkpoint selection before changing families.

If a static or harmonic model is better curve-first despite weaker scalar metrics, open a compact retraining branch around that family rather than launching a broad new model wave.

Step 3: Add Curve-First Checkpoint Selection

Update neural training infrastructure so checkpoint selection can monitor a validation curve metric, for example:

`text val_curve_mean_percentage_error_pct`

Tie-breakers should include curve P95 error, harmonic phase error, then scalar `val_mae` .

Step 4: Add Curve-Aware Losses Only After Metrics Stabilize

Do not immediately train with complex curve losses. First standardize the curve metrics. Then test a small composite loss:

`text pointwise_normalized_mse + curve_mae_term + slope_or_derivative_term + selected_harmonic_amplitude_term + selected_harmonic_phase_term`

Soft-DTW or other warp-tolerant objectives should remain secondary ablations, because physical angular phase matters for compensation.

Step 5: Preserve A TwinCAT-Friendly Path

The final target is not only an offline benchmark. The selected path must stay credible for TwinCAT/TestRig compensation.

Preferred future candidates

- sparse `RCIM` harmonic or periodic models with explicit intermediate quantities;
- curve-first selected periodic `GRU` or `LSTM` only if deployment cost and inspectability are acceptable;
- residual harmonic MLP or structured harmonic variants with harmonic-space diagnostics;
- later `Track 3` online compensation tests for uncompensated versus compensated `TE RMS` and `TE max` . Deferred exploratory candidates such as lightweight transformers, state-space sequence models, neural ODEs, and kernel/Gaussian-process baselines should stay low priority until `CVP 1.1` clarifies what the existing candidates already do on full curves.

Bottom Line

The work is no longer in a raw model-search phase. The repository now has a complete scalar and `TE Curve Verification Pipeline` evidence surface through `Wave 2.3` .

The best scalar neural result is `Wave 2.2` `periodic_gru_sequence_Bw` . The best repository static baseline remains `tree` . The best forward curve result is still paper-derived retuned `GBM` . The most important technical next step is not another large campaign, but a curve-first reranking and promotion policy that decides which existing candidates are actually suitable for continuous TE compensation.